REINFORCEMENT LEARNING

SUBJECT CODE: 24DS742

Sem-end project Synopsis

Submitted by

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Title:

A Contextual Bandit Framework for Calibrated and Cost-Aware Query Routing in Large Language Models

MASTER OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING



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JULY 2026

Abstract:

Large language models demonstrate that size and cost vary significantly. Moreover, the findings suggest that a small model could answer many questions correctly at a fraction of the cost of a large one. Furthermore, the evidence indicates that a router could send each question to the cheapest model. Additionally, this approach may save a significant amount of compute. Research shows existing routers trained with supervision or bandits. Nevertheless, the results suggest that these routers rarely examine whether their confidence appears calibrated.

However, calibration may indicate that a predicted 80% chance of success appears right about 80% of the time. Given that routers lack calibration, the evidence suggests that a router cannot reliably stay within a cost budget. Therefore, this project may propose a contextual bandit router. Recovery occurs after baseline sentence. Moreover, the important findings indicate that the router demonstrates training with a reward that combines answer correctness. Furthermore, the evidence suggests that model cost and a Brier-score term could reward honest confidence. Additionally, the router may use a small sentence encoder. Nevertheless, the results indicate that the model learns only from the outcome of the model it chooses.

Study shows deployment reflects this approach. However, the project may demonstrate that every part appears open source. Given that two Qwen2.5 models provide the foundation, the evidence suggests that the MMLU, ARC and GSM8K question sets appear relevant.

Furthermore, the findings indicate that the MiniLM encoder could support this work. In light of these results, the router may demonstrate comparison with supervised and bandit baselines. Comparison examines accuracy, cost, calibration error, budget adherence. Moreover, the important evidence suggests that the router could include a test on question types.

Nevertheless, the findings indicate that these question types may not appear during training.

Introduction:

LLMs are now used in chatbots, coding assistants and search tools. Moreover, the results may indicate that running a large model for every query appears expensive: a 7-billion-parameter model might demonstrate that it needs far more GPU memory and time than a 0.5-billion-parameter one. Furthermore, the findings could suggest that for many easy questions the small model provides the same correct answer. Given that this evidence demonstrates efficiency gains, LLM routing uses this important fact. Research shows light router examines query before answer and selects model [1]–[3].

1.1 LLM routing:

A router may learn from the past data that the small model could handle certain queries. Moreover, most routers appear to demonstrate that supervised learning provides the necessary training approach, which might indicate that this method requires the answer of every model on every training query [1], [2]. Furthermore, the deployed router could suggest that it only sees how the model it selected actually performed. Given that this partial-feedback setting presents these constraints, the results may indicate that the contextual bandit problem appears relevant. Router learns setting matches problem exactly.

1.2 Contextual bandits:

The contextual bandit approach demonstrates that an agent may observe a context, which in this case appears to be the query. Moreover, the agent could select one action, which in this case may be a model, and the agent receives a reward for that specific action only.
Furthermore, the findings suggest that this represents the simplest form of reinforcement learning, which appears to be a single step with no state transitions. Nevertheless, the standard methods may include LinUCB [4], which could maintain a linear reward estimate with an upper confidence bound. Standard methods include policy-gradient methods such as REINFORCE [5].

1.3 Calibration:

A model appears calibrated when the confidence may match how often the model is right. However, modern neural networks might demonstrate over-confidence [6]. Moreover, for a router this could matter because the routing threshold is set from the predicted probabilities: given that these probabilities are wrong, the expected cost and the expected accuracy may be wrong too. Furthermore, the Brier score [7], the mean squared gap between a predicted probability and the actual 0/1 outcome, appears to be a strictly proper scoring rule [8]. Brier score is lowest when probabilities are honest. Nevertheless, recent work could show that adding a Brier term to an RL reward may train language models to be both accurate and calibrated [9]. Additionally, this project might apply the same important idea to routing.

2. Problem Statement and Objectives:

Problem statement. Given a small LLM and a large LLM with different costs, learn a router that decides, for each incoming query, which model should answer it, using only the feedback of the chosen model, so that (i) overall accuracy stays close to always using the large model, (ii) total cost stays within a given budget, and (iii) the router's estimate of the chance that the small model is correct is calibrated.

Objectives.

- **O1.** Build an open outcome dataset by running Qwen2.5-0.5B-Instruct and Qwen2.5-7B-Instruct on MMLU, ARC and GSM8K, recording correctness and cost for every question.
- **O2.** Model routing as a two-armed contextual bandit and design a reward that combines correctness, cost and a Brier-score calibration term.
- **O3.** Train a light router (MiniLM features with a small policy head) using REINFORCE, and compare it with LinUCB, supervised and post-hoc calibrated baselines.
- **O4.** Evaluate the accuracy–cost trade-off, calibration (ECE, Brier score, reliability diagrams) and budget adherence, both on seen and on unseen question types.
- **O5.** Release all code, data and results under open-source licenses.

3. Brief Literature Review:

Table 1. Comparison to previous projects.

Title	Methodology	Results	Gaps
Hybrid LLM: Cost-Efficient and Quality-Aware Query Routing Ding et al., ICLR 2024 [2]	A BERT-style encoder predicts the quality gap between a small and a large LLM for each query; a threshold, tunable at test time, picks the route.	Up to 40% fewer calls to the large model with no drop in response quality.	Supervised: needs both models' outputs for every training query. Calibration of scores not studied.
AutoMix: Automatically Mixing Language Models Aggarwal et al., NeurIPS 2024 [24]	The small LM answers first; few-shot self-verification estimates if the answer is right; a POMDP-based router decides whether to escalate to a larger LM.	Over 50% lower compute cost at comparable performance across five datasets.	Small model must always run first; self-verification is noisy and prompt-based, not learned from reward feedback.
RouterBench: A Benchmark for Multi-LLM Routing System Hu et al., 2024 [10]	Benchmark of 405k inference outcomes from 11 LLMs on 8 datasets; evaluates KNN and MLP predictive routers and cascades within a theoretical framework.	KNN/MLP routers match the best single LLM at similar or lower cost but rarely beat a simple baseline; cascades degrade once judge error exceeds about 0.2.	Several models are closed (e.g., GPT-4, Claude). Routers judged on accuracy and cost only, not calibration.
RouteLLM: Learning to Route LLMs from Preference Data Ong et al., ICLR 2025 [1]	Four routers (similarity-weighted ranking, matrix factorisation, BERT and Llama-3-8B classifiers) trained on about 65k Chatbot Arena preference pairs plus augmentation; routes between GPT-4 and Mixtral-8x7B.	Over 2× cost reduction without loss in quality; routers transfer to model pairs not seen in training.	Needs large preference datasets and GPT-4-judged augmentation; two-model routing only; calibration not examined.
MixLLM: Dynamic Routing in Mixed Large Language Models Wang et al., NAACL 2025 [11]	Contextual-bandit-based routing with tag-enhanced query embeddings, light quality and cost predictors per LLM, a meta decision maker and continual learning.	About 97% of GPT-4's quality at about 24% of its cost under a time constraint.	Predicted quality scores are not checked for calibration; relies on closed API models.
Adaptive LLM Routing under Budget Constraints (PILOT) Panda et al., EMNLP Findings 2025 [12]	Extends LinUCB with a prior learned from human preferences in a shared query–LLM embedding space; an online multi-choice knapsack policy enforces the budget.	On RouterBench, about 93% of GPT-4's performance at about 25% of its cost; beats LinUCB and other bandit baselines.	Budget enforced only after the learning phase; needs preference data; calibration not studied.
ParetoBandit: Budget-Paced Adaptive Routing for Non-	LinUCB on MiniLM + PCA features with a primal-dual budget pacer, geometric forgetting and a hot-swap	Stays within 0.4% of the budget; +0.071 reward after a price drop; adds a new	Calibration of quality predictions not analysed; rewards come from an LLM judge; offline simulation only.

Stationary LLM Serving Taberner-Miller, 2026 [14]	model registry; three models from Llama-3.1-8B to Gemini-2.5-Pro.	model in about 142 steps; 96.4% of oracle reward.	
Learning to Route LLMs with Confidence Tokens (Self-REF) Chuang et al., 2024 [15]	LoRA fine-tunes the local LLM (Llama3-8B, Mistral-7B) to add a confidence token to each answer; low-confidence queries go to Llama3-70B.	Matches 70B accuracy while routing only 39–65% of queries (Llama3-8B); up to 2.03× lower latency.	Requires fine-tuning the LLM itself; better calibration did not always give better routing; no cost-aware learned policy.
UCCI: Calibrated Uncertainty for Cost-Optimal LLM Cascade Routing Kotte, 2026 [16]	The small model's token-margin uncertainty is mapped to an error probability with isotonic regression; the threshold minimises cost under an accuracy target (4B → 12B).	ECE reduced from 0.12 to 0.03; 31% cost saving at micro-F1 0.91 compared with the large model alone.	Supervised and post-hoc; needs white-box token probabilities; one private NER dataset; static under distribution shift.
Beyond Binary Rewards: Training LMs to Reason About Their Uncertainty (RLCR) Damani et al., ICLR 2026 [9]	RL reward = binary correctness + Brier score on the model's stated confidence; trains Qwen2.5-7B on HotpotQA and math.	HotpotQA ECE falls from 0.37 (standard RL) to 0.03 with similar accuracy (62.1% vs 63.0%); out-of-domain ECE 0.46 → 0.21.	Not applied to routing or model selection; needs costly RL training of a 7B LLM.

4. Proposed Methodology:

4.1 System overview:

The system has two phases (Fig. 1). In Phase A, both LLMs answer every question once and the outcomes are saved. In Phase B, the router is trained and tested by replaying this log. The router only sees the outcome of the model it chose, which simulates real deployment, and this phase runs entirely on a laptop CPU.

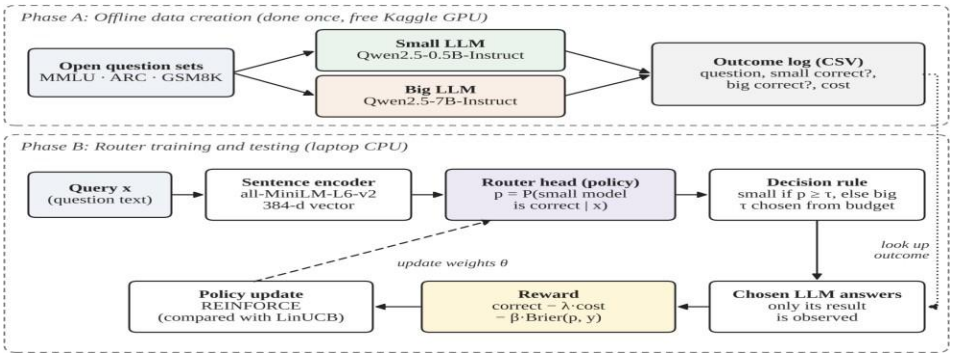


Fig. 1. Block diagram of the proposed system.

4.2 Data creation (Phase A):

The study examines that three open question sets are used: MMLU [18] (14,042 test questions over 57 subjects), ARC [19] (3,548 science questions) and GSM8K [20] (1,319 math word problems). Moreover, the small and large models may represent Qwen2.5-0.5B-Instruct and

Qwen2.5-7B-Instruct [21]. Furthermore, the findings indicate that for multiple-choice questions, a model's answer is the option with the highest log-likelihood, which could avoid slow text generation. However, the analysis suggests that for GSM8K, the model might write a solution with greedy decoding and its final number is compared with the reference. Additionally, the runs could use free Kaggle GPUs, with 4-bit loading for the 7B model if memory appears short. Cost shows model metrics cS and cB measured as average GPU time per query, scaled so cB = 1.

4.3 Bandit formulation and reward:

Given that context demonstrates x, the 384-dimensional embedding of the query from all-MiniLM-L6-v2 [22], [23], the results indicate that actions are $a \in \{S, B\}$ (small or big model). Nevertheless, the router may represent a small neural network $f\theta$ that could output $p = \sigma(f\theta(x))$, the estimated chance that the small model answers correctly. In light of training findings, the router might pick the small model with probability $\pi\theta(S \mid x) = \sigma((p - \tau)/T)$, where τ appears to be a threshold and T a temperature that controls exploration. Reward shows one query calculated as:

$$r = 1[\text{chosen model is correct}] - \lambda \cdot ca - \beta \cdot w \cdot (p - yS)2$$

Moreover, the evidence suggests that λ sets how much cost matters, β sets how much honest confidence matters, and $yS \in \{0, 1\}$ could record whether the small model was correct. Furthermore, the last term may represent the Brier score. However, the findings indicate that it is used only when the small model was chosen, because only then is yS observed, and it might be weighted by $w = 1/\pi\theta(S \mid x)$ (inverse propensity weighting, clipped) so that rarely chosen queries are not ignored. Router shows update with REINFORCE [5] using moving-average baseline b: $\theta \leftarrow \theta + \alpha(r - b)\nabla\theta \log \pi\theta(a \mid x)$, together with direct gradient of Brier term. Given that training demonstrates Adam optimiser use, the results could indicate that a few passes over the shuffled training log are conducted.

4.4 Budget-aware routing at test time:

Notwithstanding exploration absence at test time, the query may go to the small model if $p \geq \tau$, otherwise to the big one. Nevertheless, the findings suggest that given a budget, such as at most 30% of queries sent to the big model, τ could be chosen on the validation set to meet it. Moreover, the evidence indicates that because p is calibrated, the expected accuracy might be predicted before deployment as the average p of queries sent to the small model plus the big model's accuracy on the rest. Comparing shows prediction with measured accuracy tests budget-reliability of router.

4.5 Baselines:

Furthermore, the study establishes that (B1) always small; (B2) always big; (B3) random routing at the same big-model rate; (B4) a supervised router could be trained with cross-entropy on full labels, in the style of RouteLLM [1]. In light of these results, (B5) may represent B4

followed by isotonic calibration [17], in the style of UCCI [16]. Additionally, the findings indicate that (B6) LinUCB [4], in the style of PILOT [12] without the preference prior; (B7) the proposed router with $\beta = 0$ (no Brier term), as an ablation. Results show these baselines provide comparison metrics.

4.6 Evaluation:

However, the analysis suggests that splits include: (a) Seen questions: MMLU split 70/10/20 into train, validation and test. Moreover, the findings indicate that (b) Unseen question types: train on MMLU, test on ARC and GSM8K. Given that metrics demonstrate accuracy, cost and big-model call rate, the evidence could suggest that the accuracy–cost curve is obtained by sweeping τ and the area under it. Nevertheless, the results may include expected calibration error (ECE, 15 bins) [6], Brier score and reliability diagrams; budget violation rate and the gap between predicted and measured accuracy. Settings show every run with three random seeds reported as mean ± standard deviation.

5. Expected Outcomes

- An open outcome dataset of about 18,900 questions, each with the correctness of both models and their cost.
- A bandit-trained router that keeps accuracy close to the big model while sending a large share of queries to the small model, and that matches the supervised router on the accuracy–cost curve despite seeing only partial feedback.
- Lower calibration error than bandit routers trained without the Brier term, close to post-hoc isotonic calibration but without a separate calibration step.
- Better budget reliability, with predicted accuracy and cost close to the measured values.
- Evidence on how well calibration holds on unseen question types (ARC, GSM8K), with all code and results released openly.

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